Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		Attempt at starting from Faraday's law e.g. $\frac{BA}{t}$ (1) Clear explanation leading to Bvd – minimum $\frac{BA}{t} = \frac{Bxd}{t} =$ Bdv **OR** area in 1 s is dv (1)	2			2	1	
	(b)	(i)	Flux in **or** right pd (in 1 s) = B_2vd (1) Flux out **or** left pd (in 1 s) = B_1vd (1) Net flux change **or** pd around the loop (per s) = B_2vd - B_1vd (1) **Alternative:** $V = \frac{d}{dt}(BA) = \frac{AdB}{dt}$ (1) $V = \frac{AdB}{dx}\frac{dx}{dt}$ (1) $V = d^2\left(\frac{B_2 - B_1}{d}\right)v$ QED (1) **Award 2 marks for:** pd on right = B_2vd pd on left = B_1vd Hence B_2vd - B_1vd **Award 1 mark for:** It depends on the change in flux / B Hence ΔBvd	1	1 1		3	1	
		(ii)	B_2 - B_1 = 1.20 × 0.038 (1) Answer = 15.25 m[V] or 1.2 × 0.038 × 8.8 × 0.038 seen (1)		2		2	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Length = 4 × 3.8cm (1) Area = 2.2 mm × 2 mm (1) Resistivity equation used (0.55 mΩ) (1) $I = \frac{V}{R}$ used (1) Correct answer = 28 [A] (1) **ecf** 56 A and 111 A award 4 marks	 1 1	1 1 1		5	3	
			Question 6 total	**5**	**7**	**0**	**12**	**7**	**0**